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Game Theory

Lecture notes. Higher order beliefs

1 The email game

In this section we introduce the *email game* of Rubinstein [1989]. It will serve as our running example to illustrate higher-order beliefs, common knowledge, and the role of communication protocols.

1.1 A coordination problem with incomplete information

Two players, 1 and 2, must choose actions in $\{A, B\}$. There are two possible states of nature:

$$K = \{g, b\},$$

where g stands for a “good” state (e.g., there is a profitable opportunity for coordinated action) and b for a “bad” state (e.g., the opportunity is not there, or the risks are too high).

Payoffs are given by the following family of games, depending on the state k .

	A	B		A	B
A	2,2	-2,1	A	0,0	-2,1
B	1,-2	1,1	B	1,-2	1,1
State g			State b		

In state g the efficient outcome is (A, A) : both players get 2 instead of 1 at (B, B) . In state b , action A is strictly dominated by B for both players and (B, B) is the unique Nash equilibrium.

Ex ante, the players would like to play (A,A) in state g and (B,B) in state b . The difficulty is that the state is not common knowledge; instead it is learned through a noisy *email protocol*.

1.2 The email protocol and the type space

Nature first chooses the state $k \in \{g, b\}$:

$$\mathbb{P}(k = g) = \pi \in (0, 1), \quad \mathbb{P}(k = b) = 1 - \pi.$$

If $k = b$, no email is ever sent and the two players receive no information. If $k = g$, a computer sends an email to Player 1 with probability 1. Every time a player receives an email, she immediately sends a *reply* email to the other player (“I received your message”). Each email, including the initial message, is delivered independently with probability $1 - \varepsilon$ and lost with probability $\varepsilon \in (0, 1)$.

Thus, whenever a player receives an email, she knows that the state is g and that the other player has previously received at least one email. However, she does not know how many emails have been successfully exchanged so far.

A *run* of the protocol can be described by the number of emails sent by players before the first failure. Let $t \in \{0, 1, 2, \dots\}$ be the number of successfully delivered emails before the first failure or before the process terminates. One can show that in state g ,

$$\mathbb{P}(t = 0 \mid k = g) = \varepsilon, \quad \mathbb{P}(t = n \mid k = g) = (1 - \varepsilon)^n \varepsilon \quad \text{for } n \geq 1,$$

and $\mathbb{P}(t = \infty \mid k = g) = 0$, and thus we know that the email protocol almost surely terminates.

The *type* of each player is defined as the number of emails sent by the player, as follows. Let T_i be the set of types of Player i :

$$T_1 = \{0, 1, 2, \dots\}, \quad T_2 = \{0, 1, 2, \dots\}.$$

If $t = 0$, no email is delivered and both players have type 0. If $t \geq 1$, then exactly one of the players sends the last email and has type t , while the other has type $t - 1$ or type t . More precisely:

- if t is odd, then Player 1 has type t , Player 2 has type $t - 1$;
- if t is even, then Player 2 has type t , Player 1 has type t .

Types summarize all information players have about the state and about the history of communication. Conditional on her type, each player can compute a posterior belief over (k, t, t') , and therefore over (k, T_j) , $j \neq i$.

Remark 1.1. Type 0 is ambiguous for player 2: it is consistent both with $k = b$ (no opportunity) and with $k = g$ and an immediate failure of the first email. A player of type $t \geq 1$ knows that $k = g$, but may still be uncertain whether the other player has type $t - 1$, t or $t + 1$, and so on.

1.3 Strategies and equilibria

A (pure) strategy of Player i is a function

$$\sigma_i : T_i \rightarrow \{A, B\},$$

mapping types to actions. We write $\sigma_i(t)$ for the action chosen by Player i when she has type t .

We focus on Bayesian Nash equilibria in this type space: a profile (σ_1, σ_2) such that, for each i and each type $t \in T_i$, $\sigma_i(t)$ is a best reply to $\sigma_j(\cdot)$ given Player i 's posterior beliefs.

We now analyze behavior in the email game via iterative deletion of dominated actions, type by type.

Recall that $\pi \in (0, 1)$ is the prior probability of state g and that $\varepsilon \in (0, 1)$ is the failure probability of each individual email. For a player 2 of type 0, Bayes' rule gives the posterior

$$\rho := \mathbb{P}(k = g \mid T_2 = 0) = \frac{\pi\varepsilon}{(1 - \pi) + \pi\varepsilon}.$$

For a player of type $t \geq 1$, the state is known to be g with probability 1, and the other player's type is either $t - 1$ or $t + 1$ with probabilities.

$$\begin{aligned} \alpha &:= \mathbb{P}(T_2 = t - 1 \mid T_1 = t) = \mathbb{P}(T_1 = t \mid T_2 = t) = \frac{1}{2 - \varepsilon}, \\ 1 - \alpha &= \mathbb{P}(T_2 = t \mid T_1 = t) = \mathbb{P}(T_2 = t \mid T_1 = t) = \frac{1 - \varepsilon}{2 - \varepsilon}. \end{aligned}$$

The first line corresponds to the probability of either of the players sending an email which is lost, while the second one corresponds to the probability of either of the players sending an email which is well received, but whose corresponding reply is lost.

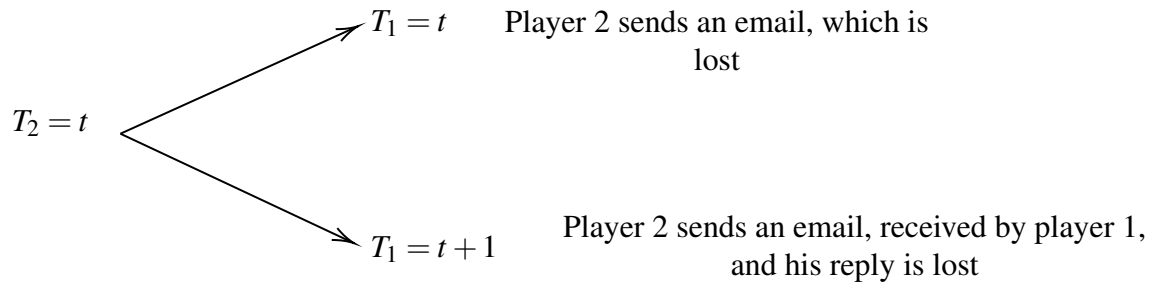
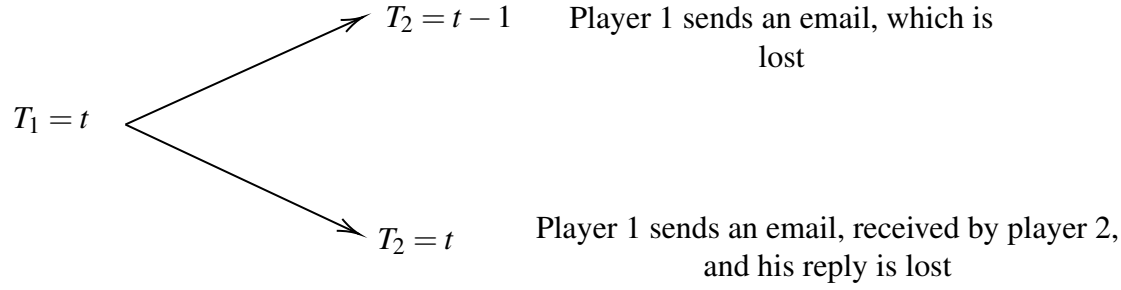
We impose the following parameter condition:

$$\rho < \frac{1}{2} \tag{1}$$

The condition can be written in terms of the primitives (π, ε) , as:

$$\frac{\pi}{1 - \pi} \geq \varepsilon$$

using the expression of ρ above.



Proposition 1.2 (Email game: iterative elimination and uniqueness). *Assume (1). Then, in the email game:*

1. *For each player and each type $t \in \{0, 1, 2, \dots\}$, action A is iteratively strictly dominated by B (in the usual sense of iterative deletion of strictly dominated actions in the type space).*
2. *In particular, in every Bayesian Nash equilibrium, both players play B at every type:*

$$\sigma_i(t) = B \quad \text{for all } i \in \{1, 2\}, t \in \{0, 1, 2, \dots\}.$$

We prove the result by going through types in increasing order, and at each step showing that, once A has been eliminated at all lower types, it is strictly dominated at the next type as well.

Lemma 1.3 (Type 0). *Under (1), for each player i at type $T_i = 0$, action A is strictly dominated by B, independently of the opponent's strategy.*

Proof. At type 0, the first player knows for sure the state is b , so its optimal strategy is playing B , as B is the dominant strategy for both players in state b .

At type 0 the second player knows the state may be g or b . Let $a_0 \in \{A, B\}$ be the action chosen by the opponent.

Case 1: $a_0 = B$. If the opponent plays B , the second player's expected payoffs from A and B are

$$\mathbb{E}[u_2(B, A) \mid T_2 = 0] = \rho \cdot (-2) + (1 - \rho) \cdot (-2), \quad \mathbb{E}[u_2(B, B) \mid T_2 = 0] = \rho \cdot 1 + (1 - \rho) \cdot 1,$$

so

$$\mathbb{E}[u_i(B, B) - u_i(A, B) \mid T_i = 0] = 3 > 0.$$

Thus B is strictly better than A when the opponent plays B .

Case 2: $a_0 = A$. If the opponent plays A , then

$$\mathbb{E}[u_2(A, A) \mid T_i = 0] = \rho \cdot 2 + (1 - \rho) \cdot 0, \quad \mathbb{E}[u_2(A, B) \mid T_2 = 0] = \rho \cdot (1) + (1 - \rho) \cdot 1.$$

The expected payoff difference is

$$\mathbb{E}[u_i(B, A) - u_i(A, A) \mid T_i = 0] = 1 - 2\rho.$$

Under the parameter restriction $\rho < 1/2$ in (1), we have $1 - 2\rho > 0$. Hence B is strictly better than A also when the opponent plays A .

Since these are the only possible actions of the opponent at type 0, we have shown that B yields a strictly higher payoff than A against any strategy of the opponent. Thus A is strictly dominated by B at type 0. $\square$

By iterative deletion, we may now restrict attention to profiles in which both players choose B at type 0.

Lemma 1.4 (Types $t \geq 1$). *Assume that, for some $t \geq 1$, all players choose B at all types $s < t$ (i.e., A has been eliminated at types $0, \dots, t-1$). Then at type t action A is strictly dominated by B (against the restricted set of opponents' strategies consistent with the previous eliminations).*

Proof. We will begin with fixing player's 1 type as $T_1 = t \geq 1$. By construction of the email protocol, she knows that $k = g$ with probability 1. Moreover, by the geometry of the protocol, her belief about the opponent's type T_2 satisfies

$$\mathbb{P}(T_2 = t - 1 \mid T_1 = t) = \alpha, \quad \mathbb{P}(T_2 = t \mid T_1 = t) = 1 - \alpha,$$

with $\alpha = 1/(2 - \varepsilon)$ as above.

Under the induction hypothesis, whenever $T_2 = t - 1$ the opponent plays B . When $T_2 = t$, her action can be either A or B ; we must show that, whatever the opponent does at type t , B strictly dominates A for a type- t player.

Let $a_t \in \{A, B\}$ be the opponent's action at type t . In state g , the payoff differences between (B, a) and (A, a) are:

$$u_1(B, B; g) - u_i(A, B; g) = 1 - (-2) = 3, \quad u_1(B, A; g) - u_i(A, A; g) = 1 - 2 = -1.$$

Thus, for a type- t player,

$$\mathbb{E}[u_1(B, \sigma_j(T_j)) - u_i(A, \sigma_j(T_j)) \mid T_i = t] = \alpha \cdot 3 + (1 - \alpha) \cdot \delta(a_t),$$

where

$$\delta(a_t) := \begin{cases} 3 & \text{if } a_t = B, \\ -1 & \text{if } a_t = A. \end{cases}$$

The “worst” case for B relative to A is when $a_t = A$, because then $\delta(a_t) = -1$. In that case the expected payoff difference simplifies to

$$3\alpha + (1 - \alpha) \cdot (-1) = 4\alpha - 1.$$

Hence a sufficient condition for B to strictly dominate A at type t is

$$4\alpha - 1 > 0 \iff \alpha > \frac{1}{4}.$$

Since $\alpha = 1/(2 - \varepsilon)$, this is always true. The crucial argument is that, from any player perspective, when she does not receive an email, it is always more likely that her email was lost rather than her email was successfully sent but the reply was lost.

So, even in the case $a_t = A$ (which is the most favorable to action A), B yields a strictly higher expected payoff, and a fortiori this holds if $a_t = B$. Thus, given that the opponent is constrained to play B at all types $< t$, action B strictly dominates A at type t for player 1.

For player 2, the analysis is the same. If player 2 is of type t , meaning $T_2 = t$, he knows the type t player 1 will optimally play B (by the argument above).

She faces a symmetric uncertainty about if player 1 is of type t (with probability α , when she knows the opponent plays b), or if the player is of type $t + 1$ (with probability $1 - \alpha$, when she does not know his action).

Analogous as before, we show that no matter the action of the player 1 of type $t + 1$, the player 2 of type t optimally plays B . $\square$

Proof of Proposition 1.2. By Lemma 1.3, under the condition $\rho < 1/2$ the action A is strictly dominated by B at type 0 for each player. After one round of elimination of strictly dominated actions, we may therefore restrict to strategies in which both players play B at type 0.

Assume now that after t rounds of elimination, for some $t \geq 0$, all players play B at all types $s \leq t$. By Lemma 1.4, the action A is then strictly dominated by B at type $t + 1$ for both players, given the restricted set of opponents’ strategies. Hence A is also eliminated at type $t + 1$ at the next round.

By induction on t , this argument shows that for each type t and each player, A is eliminated in finitely many steps of iterated deletion of strictly dominated actions. The only surviving strategies are those that play B at every type.

Any Bayesian Nash equilibrium must assign positive probability only to strategies that survive iterative deletion of strictly dominated actions; hence in every Bayesian Nash equilibrium both players choose B at all types. $\square$

Remark 1.5 (Email game and interim rationalizability). *The proof above is explicitly an iterative deletion argument, carried out in the type space. Each step eliminates A at one more type, using the fact that lower types already play B for sure and that types differ only by one unit. This is the same spirit as interim (or conditional) rationalizability: at each “round”, a type retains only actions that are best responses to some belief over opponents’ types and actions that survived the previous rounds.*

In later sections we will introduce interim correlated rationalizability (ICR) formally and show how it can be characterized on universal type spaces. The email game will serve as a concrete example where iterated reasoning about beliefs and deviations leads to a unique, seemingly pessimistic outcome (B at all types) even in a state where mutual coordination on A would be efficient.

2 The Universal Type Space

In this section we construct a space of types that contains, in a suitable sense, *all* possible hierarchies of beliefs about the state of nature and about other players’ beliefs. The construction follows the presentation made in [Maschler, Solan, and Zamir, 2013, Chapter 11] of the seminal work by Mertens and Zamir [1985], with notation adapted to our running example: K is a (fixed) compact metric space of states of nature, and $N = \{1, \dots, n\}$ is the set of players.

2.1 Belief hierarchies and consistency

A central issue in games with incomplete information is that players have beliefs about K , beliefs about other players’ beliefs about K , beliefs about other players’ beliefs about other players’ beliefs about K , and so on. In principle this leads to an infinite hierarchy:

beliefs, beliefs about beliefs, beliefs about beliefs about beliefs, ...

The following example shows that specifying only first-order beliefs (about K) is not enough to determine optimal actions in general.

Example 2.1 (Coordination depends on higher-order beliefs). Consider the simple coordination situation of the email game in which each player must decide whether to take a risky coordinated action. Even if Player 1 knows that $k = g$, she must still take into account whether Player 2 knows that $k = g$, and whether Player 2 knows that she knows that $k = g$, and so on. As we have seen, finite levels of “mutual knowledge” never amount to full common knowledge: for every finite level, there remains a small probability of misalignment, which prevents the risky coordinated action from being optimal.

Formally, fix a player i . A *hierarchy of beliefs* for i consists of a sequence

$$(\mu_i^0, \mu_i^1, \mu_i^2, \dots),$$

where

- $\mu_i^0 \in \Delta(K)$ is a belief over the state. Let us denote the space of such beliefs as $Z_i^{(1)}$
- μ_i^1 is a belief over $(K \times Z_{-i}^{(1)})$, or equivalently a belief over $(K \times Z_1^{(1)} \times \cdots \times Z_{i-1}^{(1)} \times Z_{i+1}^{(1)} \times \cdots \times Z_n^{(1)})$, or $\mu_i^1 \in \Delta(K \times Z_{-i}^{(1)})$. Let us denote the space of such beliefs as $Z_i^{(2)}$
- more generally, μ_i^m is a belief over $(K \times Z_{-i}^{(m)})$, where $Z_{-i}^{(m)}$ is a space of (m th-order) types of players $-i$, or $\mu_i^m \in \Delta(K \times Z_{-i}^{(m)})$

In other words, μ_i^1 describes Player i 's beliefs about the state and about the other players' types; μ_i^2 is a belief about the state and about others' beliefs about the state and types of others; and so on.

Such hierarchies must be *consistent*. Intuitively, the marginal of μ_i^{m+1} on $(K \times Z_{-i}^{(m)})$ must coincide with μ_i^m , because beliefs about $(m+1)$ st-order information should be compatible with beliefs about m th-order information. We make this precise shortly.

Example 2.2 (Inconsistent belief hierarchy (MSZ Example 11.4)). Let $N = \{1, 2\}$ and $K = \{k_1, k_2\}$. A first-order belief of player 1 is a probability distribution $\mu^1 \in \Delta(K)$, say

$$\mu^1(k_1) = p, \quad \mu^1(k_2) = 1 - p.$$

A second-order belief of player 1 is a pair (μ^1, ν^1) where $\nu^1 \in \Delta(K \times \Delta(K))$ describes player 1's beliefs about the state and about player 2's first-order belief. Suppose that ν^1 assigns probability α to (k_1, q) and $1 - \alpha$ to (k_2, r) , where $q, r \in \Delta(K)$ are the (possible) first-order beliefs of player 2. The marginal of ν^1 on K is then

$$\nu^1(k_1) = \alpha, \quad \nu^1(k_2) = 1 - \alpha.$$

If $p \neq \alpha$ then the first-order belief μ^1 and the marginal of ν^1 on K assign different probabilities to the state k_1 . Hence (μ^1, ν^1) is *not* consistent: the same agent would give two different answers to the question “what probability do you assign to k_1 ?”

2.2 Type spaces and beliefs over types

We are now going to introduce the formalism of Harsanyi type spaces.

A *type space* consists of:

- a measurable space T_i for each player i ,
- a measurable map $\beta_i : T_i \rightarrow \Delta(K \times T_{-i})$,

where $T_{-i} = T_1 \times \cdots \times T_{i-1} \times T_{i+1} \times \cdots \times T_n$. A type $t_i \in T_i$ represents everything Player i knows and believes about K and about the other players' types, and $\beta_i(t_i)$ is the associated belief:

$$\beta_i(t_i) \in \Delta(K \times T_{-i}).$$

We write

$$\beta_i^K(t_i) \in \Delta(K)$$

for the marginal of $\beta_i(t_i)$ on K .

The consistency of beliefs across players and types is encoded by the type space itself: if the actual state and type vector is $(k, t) \in K \times T$, then Player i 's belief $\beta_i(t_i)$ must assign probability 1 to $(K \times T_{-i})$ -coordinates compatible with the actual types t_{-i} (in a precise measurability sense). In this way, types “explain” the full hierarchy of conditional beliefs of each player.

Remark 2.3 (From signals to types). *Any standard Harsanyi model in which each player receives a signal (possibly from a complicated information structure) can be represented in this framework by taking T_i to be the set of possible signals of Player i , with β_i representing posterior beliefs conditional on those signals. The notion of a type space abstracts away from the generation of signals and keeps only the information relevant for beliefs and decision making.*

2.3 Consistent hierarchies and canonical types

The relation between types and belief hierarchies can be made explicit. For a type $t_i \in T_i$, define recursively:

$$\begin{aligned}\mu_i^0(t_i) &:= \beta_i^K(t_i) \in \Delta(K), \\ \mu_i^1(t_i) &:= \beta_i(t_i) \in \Delta(K \times Z_{-i}^{(1)}), \\ \mu_i^{m+1}(t_i) &:= \text{the belief over } (K \times Z_{-i}^{(m)}) \text{ induced by } \beta_i(t_i) \quad (m \geq 1).\end{aligned}$$

Here, the meaning of “induced by” is made clear in Subsection 4.2.

Thus each type gives rise to a full, consistent hierarchy of beliefs. Conversely, the universal type space constructed below contains a *canonical type* for each consistent hierarchy of beliefs, and every type in any type space can be represented as such a canonical type.

Example 2.4 (Canonical types reproduce hierarchies). Suppose we start with a consistent hierarchy $(\mu_i^0, \mu_i^1, \mu_i^2, \dots)$ for Player i . The universal type space contains a canonical type $\hat{t}_i$ such that the belief hierarchy induced by $\hat{t}_i$ is exactly $(\mu_i^0, \mu_i^1, \mu_i^2, \dots)$. In this way, the universal type space “collects” all such hierarchies in a single topological structure.

At this stage it is useful to keep in mind the following guiding principle:

Types are belief hierarchies, and the universal type space contains precisely all consistent hierarchies.

In the next subsections we construct this universal type space formally, establish its compactness and metric structure when K is compact metric, and show that it plays a “terminal” role among all type spaces: any type space admits a unique, belief-preserving projection onto the universal one.

2.4 Consistent belief hierarchies and canonical types

We now turn to the formal construction of the universal type space. Throughout, K denotes a fixed compact metric space of states of nature and $N = \{1, \dots, n\}$ the set of players.

Spaces of hierarchies

For each $m \geq 0$, let H_i^m denote the space of m th-order beliefs of Player i , defined recursively as follows:

- $H_i^0 := \Delta(K)$;
- given H_j^{m-1} for all $j \neq i$, define

$$H_i^m := \Delta\left(K \times \prod_{j \neq i} H_j^{m-1}\right).$$

Intuitively, an element $h_i^m \in H_i^m$ specifies Player i 's belief about the state and about the $(m-1)$ st-order beliefs of the other players. (Thus the space $\prod_{j \neq i} H_j^{m-1}$ plays the role of the $(m-1)$ st-order types of players $-i$.) We write

$$H_i := \prod_{m \geq 0} H_i^m$$

for the space of all (candidate) hierarchies of beliefs of Player i .

Elements of H_i are of the form

$$h_i = (h_i^0, h_i^1, h_i^2, \dots), \quad h_i^m \in H_i^m.$$

Consistency

The consistency requirement is that for each $m \geq 0$ the following diagram commutes:

Remark 2.5 (Consistency as a commutative diagram). *Let $\pi_m : H_i^{m+1} \rightarrow H_i^m$ be the marginalization map and $\text{pr}_k : H_i^{\text{cons}} \rightarrow H_i^k$ the canonical projections. Then $h_i \in H_i^{\text{cons}}$ is a consistent hierarchy iff, for every $m \geq 0$, the following triangle commutes:*

$$\begin{array}{ccc} & & H_i^{m+1} \\ & \nearrow \text{pr}_{m+1} & \downarrow \pi_m \\ H_i^{\text{cons}} & \xrightarrow{\text{pr}_m} & H_i^m \end{array}$$

that is,

$$\pi_m(\text{pr}_{m+1}(h_i)) = \text{pr}_m(h_i) \quad \text{for all } h_i \in H_i^{\text{cons}}.$$

In particular, H_i^{cons} is the inverse limit $\varprojlim_m H_i^m$ of the system (H_i^m, π_m) .

where $\pi_m : H_i^{m+1} \rightarrow H_i^m$ is marginalization on $K \times \prod_{j \neq i} H_j^{m-1}$. A hierarchy $h_i = (h_i^0, h_i^1, h_i^2, \dots)$ is then consistent if and only if

$$\pi_m(h_i^{m+1}) = h_i^m \quad \text{for all } m \geq 0,$$

that is, each higher-order belief projects onto the previous one. The space H_i^{cons} of consistent hierarchies is the inverse limit of this system:

$$H_i^{\text{cons}} = \varprojlim_m H_i^m.$$

Remark 2.6 (Non-emptiness). *Every type in any type space induces a consistent hierarchy, by iterating belief maps. Hence H_i^{cons} is nonempty and contains all hierarchies that arise from Bayesian posteriors in Harsanyi models.*

Canonical types

We now define the *canonical type space* of Player i as

$$\hat{T}_i := H_i^{\text{cons}}.$$

An element $\hat{t}_i \in \hat{T}_i$ is a consistent hierarchy

$$\hat{t}_i = (\hat{\mu}_i^0, \hat{\mu}_i^1, \hat{\mu}_i^2, \dots), \quad \hat{\mu}_i^m \in H_i^m.$$

Given $\hat{t}_i$, we define the belief

$$\hat{\beta}_i(\hat{t}_i) := \hat{\mu}_i^1 \in \Delta\left(K \times \prod_{j \neq i} \hat{T}_j\right).$$

In other words, $\hat{\mu}_i^1$ is Player i 's belief over the state and over the (canonical) types of the other players. Higher-order beliefs are already encoded in the components $\hat{\mu}_i^m$, $m \geq 2$.

Proposition 2.7 (Canonical type space). *$\hat{T} := \prod_{i=1}^n \hat{T}_i$ together with belief maps $\hat{\beta}_i : \hat{T}_i \rightarrow \Delta(K \times \hat{T}_{-i})$ is a type space (in the sense defined above). Moreover, every type in any type space induces a unique canonical type in $\hat{T}$.*

Idea of proof. Given any type t_i in a type space (T_i, β_i) we define inductively the hierarchy $(\mu_i^0, \mu_i^1, \mu_i^2, \dots)$ by

$$\mu_i^0 := \beta_i^K(t_i), \quad \mu_i^1 := \beta_i(t_i), \quad \mu_i^{m+1} := \text{the belief induced by } \mu_i^m \text{ on } K \times T_{-i}^m,$$

where T_{-i}^m denotes the (m th-order) component of types of players $-i$. These beliefs form a consistent hierarchy by construction, and the resulting canonical type $\hat{t}_i$ is unique.

Conversely, given a canonical type $\hat{t}_i$, the map $\hat{\beta}_i(\hat{t}_i) = \hat{\mu}_i^1$ is the belief over $(K \times \hat{T}_{-i})$ that generates the full hierarchy, so that $\hat{T}$ is a type space. $\square$

Remark 2.8 (Canonical representation). *Each t_i in any type space corresponds to a unique canonical type $\hat{t}_i$ whose hierarchy of beliefs coincides with that induced by t_i . Hence $\hat{T}$ “represents” all types up to belief equivalence and is called the universal (or canonical) type space.*

Guiding intuition

Every type t_i in any model encodes a consistent hierarchy of beliefs. The canonical construction simply *reveals* this hierarchy as an explicit infinite tuple of probability measures. In the universal type space, types *are* hierarchies of beliefs, and belief maps $\hat{\beta}_i$ simply extract the first-order component of these hierarchies. Higher-order beliefs are built in by the recursive structure.

2.5 The universal property and projections

So far we have constructed a space $\hat{T}$ that contains all consistent hierarchies of beliefs. We now show that $\hat{T}$ is universal in the following precise sense: any type space admits a unique *belief-preserving* projection into $\hat{T}$. This justifies calling $\hat{T}$ the *universal type space*.

Belief-preserving maps

Let $(T_i, \beta_i)_{i \in N}$ and $(T'_i, \beta'_i)_{i \in N}$ be two type spaces (with the same state space K). A *belief-preserving map* from T to T' is a family of measurable maps

$$\varphi_i : T_i \rightarrow T'_i, \quad i \in N,$$

such that the belief of t_i in T coincides with the belief of $\varphi_i(t_i)$ in T' after applying φ to the opponents' types.

Formally, for every i and $t_i \in T_i$ the following diagram commutes:

$$\begin{array}{ccc} T_i & \xrightarrow{\varphi_i} & T'_i \\ \beta_i \downarrow & & \downarrow \beta'_i \\ \Delta(K \times T_{-i}) & \xrightarrow{\text{id}_K \times \varphi_{-i}} & \Delta(K \times T'_{-i}) \end{array}$$

Remark 2.9. Two types t_i and t'_i (possibly in different type spaces) are *belief-equivalent* if they induce the same distribution over $K \times T_{-i}$. *Belief-preserving maps identify belief-equivalent types across type spaces and express formally that the belief structure is respected.*

The canonical embedding

Let (T_i, β_i) be any type space. For each player i and each type t_i , we construct its canonical hierarchy

$$\hat{\mu}_i^0 := \beta_i^K(t_i), \quad \hat{\mu}_i^1 := \beta_i(t_i), \quad \hat{\mu}_i^{m+1} := \text{the belief induced by } \hat{\mu}_i^m.$$

As we observed above, this hierarchy is consistent, hence determines a unique canonical type $\hat{t}_i \in \hat{T}_i$.

Define $\varphi_i : T_i \rightarrow \hat{T}_i$ by

$$\varphi_i(t_i) := \hat{t}_i.$$

Proposition 2.10 (Canonical projection). $\varphi = (\varphi_i)_{i \in N}$ is a belief-preserving map from T to $\hat{T}$.

Idea. By construction of the hierarchy $(\hat{\mu}_i^m)$, the first-order belief $\hat{\mu}_i^1$ is exactly $\beta_i(t_i)$ pushed forward along the canonical identification of the opponents' types. The recursive construction of higher-order beliefs makes the same identity hold at each order, hence also at order 1. $\square$

Universality

We are now ready for the central result (Maschler et al. [2013], Theorem 11.16).

Theorem 2.11 (Universal property). $\hat{T}$ is universal in the following sense: for any type space T there exists a unique belief-preserving map

$$\varphi : T \rightarrow \hat{T}.$$

In other words, among all type spaces on K , $\hat{T}$ is a terminal object with respect to morphisms that preserve beliefs.

Sketch. Existence was shown above by constructing φ_i from hierarchies. For uniqueness, suppose $\psi = (\psi_i)$ is another morphism $T \rightarrow \hat{T}$. Then for each $t_i \in T_i$, both $\varphi_i(t_i)$ and $\psi_i(t_i)$ induce the same belief hierarchy, which determines a unique canonical type. Hence $\psi_i(t_i) = \varphi_i(t_i)$ for all t_i . $\square$

Remark 2.12 (Interpretation). *Every type t_i in any game corresponds to a unique point $\varphi_i(t_i)$ in the universal type space. This canonical image contains precisely the belief information encoded by t_i , and ignores the details of the underlying signal structure. Thus two different Harsanyi models that define the same belief hierarchy give rise to the same canonical type.*

From type spaces to the UTS

The map $\varphi : T \rightarrow \hat{T}$ is called the *projection* of T onto the universal type space. It extracts exactly the epistemic content of T : types with identical belief hierarchies are mapped to the same canonical type. In fact, the canonical type space *quotients out* all irrelevant distinctions between models that do not affect players' beliefs.

In the next section we will see that this universal property plays a crucial role in epistemic game theory: interim correlated rationalizability (ICR) and other belief-based solution concepts depend only on canonical types, and are therefore independent of the specific Harsanyi model.

3 Interim Correlated Rationalizability

We now introduce *interim correlated rationalizability* (ICR). The idea is to iteratively delete actions that are never best replies to beliefs that are compatible with a player's type and with the requirement that other players also behave rationally, conditional on their types.

Throughout this section we fix:

- a compact metric state space K ;
- a finite set of players $N = \{1, \dots, n\}$;
- for each player i , a finite action set A_i and payoff function

$$u_i : K \times A \rightarrow \mathbb{R}, \quad A := \prod_{j \in N} A_j;$$

- a type space $(T_i, \beta_i)_{i \in N}$, where

$$\beta_i : T_i \rightarrow \Delta(K \times T_{-i})$$

specifies Player i 's beliefs about (K, T_{-i}) at each type t_i .

3.1 Best-reply operator and ICR

We start by defining, for each player i , an operator that assigns to every type t_i and every family of surviving actions for the other players a set of best replies.

Definition 3.1 (Best-reply correspondence br_i). Let $S_j : T_j \rightrightarrows A_j$ be correspondences assigning to each type $t_j \in T_j$ a nonempty set $S_j(t_j) \subseteq A_j$ of “currently admissible” actions for player j , and let $S_{-i} = (S_j)_{j \neq i}$.

For a given type $t_i \in T_i$, define

$$\text{br}_i(t_i; S_{-i}) \subseteq A_i$$

as the set of actions $a_i \in A_i$ for which there exists a belief

$$\mu \in \Delta(K \times T_{-i} \times A_{-i})$$

with the following properties:

1. **Consistency with t_i :** the marginal of μ on $K \times T_{-i}$ coincides with $\beta_i(t_i)$;
2. **Respect for surviving actions:** μ assigns probability 1 to action profiles a_{-i} that belong to the current admissible sets of the other players (in other words, the support is the set of surviving strategies of the opponents):

$$\mu(\{(k, t_{-i}, a_{-i}) : a_j \in S_j(t_j) \text{ for all } j \neq i\}) = 1;$$

3. **Optimality of a_i :** a_i maximizes Player i 's expected payoff against μ , i.e.

$$\mathbb{E}_\mu[u_i(k, a_i, a_{-i})] \geq \mathbb{E}_\mu[u_i(k, a'_i, a_{-i})] \quad \text{for all } a'_i \in A_i.$$

In words, $\text{br}_i(t_i; S_{-i})$ is the set of actions that are best replies at type t_i to some belief that:

- is consistent with Player i 's information and beliefs (encoded in $\beta_i(t_i)$), and
- assigns zero probability to deviations by other players to actions that have already been ruled out for their types.

We require neither independence across players nor any particular structure of μ : beliefs over $(K \times T_{-i} \times A_{-i})$ can be arbitrarily correlated, as long as they respect types and surviving actions.

Definition 3.2 (Interim correlated rationalizability (ICR)). For each player i and type $t_i \in T_i$, we define families of *surviving actions* $(A_i^m(t_i))_{m \geq 0}$ by induction:

- initial step:

$$A_i^0(t_i) := A_i \quad \text{for all } i \in N, t_i \in T_i;$$

- inductive step: given $A_{-i}^m = (A_j^m)_{j \neq i}$, define

$$A_i^{m+1}(t_i) := \text{br}_i(t_i; A_{-i}^m) \quad \text{for all } t_i \in T_i.$$

The set of *interim correlated rationalizable* actions for Player i at type t_i is

$$A_i^{\text{ICR}}(t_i) := \bigcap_{m \geq 0} A_i^m(t_i).$$

Thus an action a_i is ICR at type t_i if it survives all rounds of this iterative procedure: at each step it is a best reply to some belief that is compatible with Player i 's type and assigns probability 1 to other players using only actions that have also survived so far.

Remark 3.3 (ICR as iterated deletion of dominated actions). *The ICR procedure can be viewed as a generalized iterated deletion of strictly dominated actions in the type space. At each step $m + 1$, we delete from $A_i^m(t_i)$ all actions that are never best replies to any belief consistent with t_i and with the restriction that the other players use only actions in A_{-i}^m . This is exactly the type-by-type refinement used in the email game, now formulated in a general setting.*

3.2 ICR depends only on canonical types

We now show that interim correlated rationalizability depends only on the hierarchies of beliefs encoded in types, not on the particular Harsanyi model. In particular, it depends only on the canonical types in the universal type space.

The canonical game on the UTS

Fix a game with:

- compact metric state space K ;
- finite player set N and action sets $(A_i)_{i \in N}$;
- payoff functions $u_i : K \times A \rightarrow \mathbb{R}$;
- an arbitrary type space $(T_i, \beta_i)_{i \in N}$.

From the previous section, this type space admits a unique canonical projection $\varphi = (\varphi_i)_{i \in N}$ into the universal type space $\widehat{T} = (\widehat{T}_i, \widehat{\beta}_i)_{i \in N}$, with $\varphi_i : T_i \rightarrow \widehat{T}_i$ a morphism preserving beliefs:

$$\beta_i(t_i) = (\text{id}_K \times \varphi_{-i})_{\#} \widehat{\beta}_i(\varphi_i(t_i)) \quad \text{for all } t_i \in T_i.$$

We can therefore consider the *canonical game* on the universal type space, with the same (K, A_i, u_i) but type space $(\widehat{T}_i, \widehat{\beta}_i)$. For this canonical game we can run the same ICR procedure and obtain sets $\widehat{A}_i^m(\widehat{t}_i)$ and $\widehat{A}_i^{\text{ICR}}(\widehat{t}_i)$ for each player i , each type $\widehat{t}_i \in \widehat{T}_i$, and each round $m \geq 0$.

ICR is preserved under canonical projection

The key fact is that the canonical projection φ commutes with the best-reply operator br_i and therefore preserves the ICR sets.

Proposition 3.4 (ICR and morphisms). *Let (T_i, β_i) and (T'_i, β'_i) be two type spaces on the same (K, A_i, u_i) , and let $\psi = (\psi_i)$ be a morphism from T to T' . Then for all $i \in N$, $t_i \in T_i$ and all $m \geq 0$,*

$$A_i^m(t_i) \subseteq A_i'^m(\psi_i(t_i)),$$

where A_i^m and $A_i'^m$ are the iterates of the ICR operator on T and T' respectively. In particular,

$$A_i^{\text{ICR}}(t_i) \subseteq A_i'^{\text{ICR}}(\psi_i(t_i)).$$

If ψ is an isomorphism (bijective morphism with inverse also a morphism), then equality holds.

Idea. We argue by induction on m .

For $m = 0$, $A_i^0(t_i) = A_i'^0(\psi_i(t_i)) = A_i$ by definition.

Suppose the inclusion holds at step m for all players and types. Fix i and $t_i \in T_i$. By definition, $a_i \in A_i^{m+1}(t_i)$ if and only if there exists a belief

$$\mu \in \Delta(K \times T_{-i} \times A_{-i})$$

such that:

- the marginal of μ on $(K \times T_{-i})$ is $\beta_i(t_i)$;
- μ assigns probability 1 to action profiles a_{-i} with $a_j \in A_j^m(t_j)$ for all $j \neq i$;
- a_i is optimal against μ .

Using the morphism property of ψ , we can push μ forward along $(\text{id}_K \times \psi_{-i})$ to obtain a belief

$$\mu' \in \Delta(K \times T'_{-i} \times A_{-i})$$

whose marginal on $K \times T'_{-i}$ is $\beta'_i(\psi_i(t_i))$. By the induction hypothesis, the support of μ is contained in $A_{-i}^m \times T_{-i}$, hence the support of μ' is contained in $A_{-i}^m \times T'_{-i}$. Moreover, the expected payoff of a_i against μ' coincides with its payoff against μ . Thus a_i is also a best reply at type $\psi_i(t_i)$ to an admissible belief in the T' -game, so $a_i \in A_i^{m+1}(\psi_i(t_i))$.

This proves the inclusion at step $m+1$. If ψ is an isomorphism, the same argument applied to ψ^{-1} shows the reverse inclusion, hence equality at all steps and therefore at the ICR limit. $\square$

Applying this proposition to the canonical projection $\varphi : T \rightarrow \hat{T}$ yields the following corollary.

Corollary 3.5 (UTS sufficiency for ICR). *Let (T_i, β_i) be any type space and let $\varphi : T \rightarrow \hat{T}$ be its canonical projection into the universal type space. Then for every player i , every type $t_i \in T_i$ and every $m \geq 0$,*

$$A_i^m(t_i) \subseteq \hat{A}_i^m(\varphi_i(t_i)),$$

and hence

$$A_i^{\text{ICR}}(t_i) \subseteq \hat{A}_i^{\text{ICR}}(\varphi_i(t_i)).$$

If two types t_i, t'_i (possibly in different type spaces) have the same canonical image $\varphi_i(t_i) = \varphi_i(t'_i)$, then

$$A_i^{\text{ICR}}(t_i) = A_i^{\text{ICR}}(t'_i).$$

Remark 3.6 (Only canonical types matter). *Corollary 3.5 formalizes the idea that interim correlated rationalizability depends only on the canonical types: if two types induce the same belief hierarchy, then they admit exactly the same ICR actions in every game with the same (K, A_i, u_i) . The universal type space therefore contains precisely the information needed to compute ICR in any game.*

3.3 Strictly proper scoring rules on $\Delta(K)$

In this subsection we show that for any compact metric state space K there exists a continuous payoff function

$$s : \Delta(K) \times K \rightarrow \mathbb{R}$$

that *elicits beliefs truthfully*. Such a function is called a *strictly proper scoring rule*. It will be used in the next section to construct a single “universal revealing game” in which the unique ICR-surviving action of a type is to announce its canonical type.

Definition

Let K be a compact metric space and let $\Delta(K)$ denote the set of Borel probability measures on K with the weak* topology (defined at Subsection 4.1).

Definition 3.7 (Strictly proper scoring rule). A continuous function $s : \Delta(K) \times K \rightarrow \mathbb{R}$ is a *strictly proper scoring rule* if, for every $p \in \Delta(K)$,

$$\arg \max_{q \in \Delta(K)} \mathbb{E}_{k \sim p}[s(q, k)] = \{p\}.$$

Equivalently, the expected score $\mathbb{E}_p[s(q, k)]$ is uniquely maximized at $q = p$.

In words, a strictly proper scoring rule is a payoff function that rewards a player for reporting a distribution q close to the true distribution p , and achieves its maximum only when $q = p$. Such functions are often used in the econometrics and machine-learning literature to elicit forecasts and probability assessments.

Existence theorem

We now state and prove a general existence result for strictly proper scoring rules. The proof uses the compactness and metrizable of K and the fact that the space $C(K)$ of continuous real-valued functions on K separates points and is dense in all relevant spaces.

Theorem 3.8 (Existence). *If K is compact metric, then there exists a continuous strictly proper scoring rule $s : \Delta(K) \times K \rightarrow \mathbb{R}$.*

Proof. Consider the (separable) Banach space $C(K)$ with the sup norm. It is separable as a result of the Stone-Weierstrass theorem. Since K is compact metric, the Riesz representation theorem identifies $\Delta(K)$ with a compact subset of the dual space $C(K)^*$ via

$$p \mapsto \left(\varphi \mapsto \int_K \varphi(k) dp(k) \right).$$

Let $(\varphi_n)_{n \geq 1}$ be a countable dense subset of the unit ball of $C(K)$; density means that for any $\varphi \in C(K)$ and any $\varepsilon > 0$ there exists n such that $\|\varphi - \varphi_n\|_\infty < \varepsilon$.

For each n , define

$$\Phi_n(q, k) := \varphi_n(k) - \mathbb{E}_q[\varphi_n],$$

and set

$$s(q, k) := - \sum_{n \geq 1} 2^{-n} \Phi_n(q, k)^2.$$

Since each Φ_n is continuous in (q, k) , the sum converges uniformly and defines a continuous function on $\Delta(K) \times K$.

Fix $p \in \Delta(K)$ and define

$$L(q) := \mathbb{E}_p[s(q, k)].$$

Then

$$L(q) - L(p) = - \sum_{n \geq 1} 2^{-n} (\mathbb{E}_p[\varphi_n] - \mathbb{E}_q[\varphi_n])^2 \leq 0,$$

with equality if and only if $\mathbb{E}_p[\varphi_n] = \mathbb{E}_q[\varphi_n]$ for all $n \geq 1$. By density of (φ_n) , these equalities hold if and only if $\mathbb{E}_p[\varphi] = \mathbb{E}_q[\varphi]$ for all $\varphi \in C(K)$, which by the Riesz representation theorem holds if and only if $p = q$. Thus $L(q)$ is uniquely maximized at $q = p$, as desired. $\square$

Remark 3.9 (Countable family of separating functions). *The construction above uses a countable dense family (φ_n) in $C(K)$. Although the scoring rule s involves an infinite series, the resulting payoff function is continuous and finite everywhere. Each term in the sum penalizes a deviation of the report q from the true distribution p along one direction φ_n , and the whole function separates points of $\Delta(K)$.*

Remark 3.10 (Strict vs. proper). *Strict propriety means that $q = p$ is the unique maximizer of the expected score under p . One can also define non-strict proper scoring rules in which p is only a maximizer; we do not need this distinction here.*

3.4 The universal revealing game

We now combine the previous sections to construct a single game in which each player has exactly one ICR action: to reveal her canonical type. This shows that the universal type space contains precisely the information needed to play the ICR procedure in any game.

Formal construction

Fix K compact metric. Consider the (static) game with:

- player set N ;
- state space K ;
- type space the universal type space $(\widehat{T}_i, \widehat{\beta}_i)$;
- for each player i , action set $A_i := \Delta(K)$, interpreted as reports of beliefs over K ;
- payoff functions determined by a strictly proper scoring rule s :

$$u_i(k, a_i, a_{-i}) := s(a_i, k) \quad \text{for all } i \in N, k \in K, a \in A.$$

Thus each player reports a probability distribution $a_i \in \Delta(K)$ about the state k , and receives payoff $s(a_i, k)$.

Messages as reports and types as beliefs

By definition of the canonical type space, each type $\hat{t}_i$ induces a first-order belief

$$\hat{\mu}_i^0 = \hat{\beta}_i^K(\hat{t}_i) \in \Delta(K).$$

Hence $\hat{t}_i$ endows Player i with a well-defined belief about the state. The action $a_i := \hat{\mu}_i^0$ is the action “report your belief” in this game. We now show that, under ICR, this action uniquely survives.

ICR chooses truthful revelation

Let br_i denote the best-reply operator in this game. Fix a player i and a canonical type $\hat{t}_i$. Consider the belief

$$\mu_i(k, \hat{t}_{-i}, a_{-i}) := \hat{\beta}_i(\hat{t}_i)(k, \hat{t}_{-i}) \cdot \delta_{a_{-i} = \hat{\mu}_{-i}^0},$$

which assigns probability 1 to opponents’ reporting their own belief $\hat{\mu}_{-i}^0$. Then for the action $a_i = \hat{\mu}_i^0$,

$$\mathbb{E}_{\mu_i}[u_i(k, a_i, a_{-i})] = \mathbb{E}_{k \sim \hat{\beta}_i^K(\hat{t}_i)} s(\hat{\mu}_i^0, k),$$

which by strict propriety of s is uniquely maximized at $a_i = \hat{\mu}_i^0$. Hence $\hat{\mu}_i^0 \in \text{br}_i(\hat{t}_i; S_{-i})$ for any admissible S_{-i} containing $\hat{\mu}_{-i}^0$.

Conversely, suppose $a_i \neq \hat{\mu}_i^0$. By the defining property of s , $\hat{\mu}_i^0$ strictly dominates a_i with respect to $\hat{t}_i$ in a one-shot sense: for every belief about (k, a_{-i}) consistent with $\hat{t}_i$, the expected score of $\hat{\mu}_i^0$ is strictly larger than that of a_i . Therefore a_i is eliminated in the first step of the ICR procedure, and never re-enters.

Proposition 3.11 (ICR reveals canonical types). *In the universal revealing game,*

$$A_i^{\text{ICR}}(\hat{t}_i) = \{\hat{\mu}_i^0\} \quad \text{for every } i \in N \text{ and } \hat{t}_i \in \hat{T}_i.$$

Proof. The argument above shows that $\hat{\mu}_i^0$ is admissible at every round of ICR (because it is a best reply to itself), whereas any $a_i \neq \hat{\mu}_i^0$ is eliminated immediately by strict propriety of s . Thus the only action that survives at $\hat{t}_i$ is $\hat{\mu}_i^0$. $\square$

Remark 3.12 (Types reconstructed). *Since $\hat{\mu}_i^0$ determines the first-order beliefs of $\hat{t}_i$ and the canonical type $\hat{t}_i$ encodes all higher-order beliefs, this game extracts exactly the epistemic information represented by the canonical type. Higher-order beliefs are indirectly revealed through the fact that $\hat{T}$ itself contains them as types.*

Remark 3.13 (Universal content of ICR). *Combining Corollary 3.5 and Proposition 3.11 yields the following interpretation:*

the universal type space coincides with the set of types that survive ICR in the universal revealing game.

Thus canonical types are exactly the epistemically rationalizable types in a single constructing game.

This completes the proof of the claim announced at the beginning of this section: the universal type space contains precisely the information needed to compute ICR in any game, and ICR itself “recovers” types in a single game via truthful revelation of beliefs.

3.5 Conclusion

The universal type space collects all possible consistent hierarchies of beliefs in a single compact metric structure. Interim correlated rationalizability then refines players’ actions conditional on their types by the iterative elimination of strategies that are never optimal under beliefs compatible with those types. The two ideas combine: types are hierarchies of beliefs and ICR uses precisely such hierarchies to define rationalizable behavior.

Strictly proper scoring rules provide a continuous mechanism to elicit beliefs, and the universal revealing game shows that the universal type space itself is the epistemic content extracted by ICR in a single game: canonical types survive and reveal exactly their belief hierarchies. In this sense, the universal type space is the epistemic foundation of ICR in arbitrary games.

These results underpin much of modern epistemic game theory: further refinements, solution concepts, and comparative statics can be understood directly on the universal type space without reference to particular signal structures.

4 Mathematical notes

4.1 The weak* topology

For any normed vector space X , we define its dual as $X' = \mathcal{L}(X, \mathbb{K})$ the space of linear continuous functions from X to $\mathbb{K}$ (where $\mathbb{K}$ is equal to either $\mathbb{R}$ or $\mathbb{C}$).

For every $x \in X$ we define the linear functionals ev_x on X' by:

$$\text{ev}_x : X' \rightarrow \mathbb{K}, \quad \text{ev}_x(f) = f(x).$$

Definition 4.1. The weak* topology on X' is defined as the coarsest topology on X' that keeps the maps $\{\text{ev}_x\}_{x \in X}$ continuous.

Convergence under the weak* topology is characterised by:

Proposition 4.2. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in X' and $f \in X'$. Then, f_n converges to f under the weak* topology if and only if $f_n(x) \rightarrow f(x)$, for all $x \in X$

4.2 The pushforward measure and higher order induced beliefs

Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces and let $f : X \rightarrow Y$ be a measurable map. For any probability measure $\mu \in \Delta(X)$, the *pushforward* measure $f_{\#}\mu \in \Delta(Y)$ is defined by

$$(f_{\#}\mu)(B) := \mu(f^{-1}(B)) \quad \text{for all } B \in \mathcal{G}.$$

We say that $f_{\#}\mu$ is the belief *induced* by μ through the map f .

We now explain how this notion is used to define higher-order belief hierarchies as in Section 2.3

Fix a type profile $(t_i, \beta_i)_{i \in I}$.

First-order belief. Define the first-order belief mapping

$$\beta_i^1 : T_i \rightarrow \Delta(K)$$

by

$$\beta_i^1(t_i) := (\pi^1)_{\# \beta_i(t_i)},$$

where $\beta_i(t_i) \in \Delta(K \times T_{-i})$ is player i 's belief at type t_i and

$$\pi^1 : K \times T_{-i} \rightarrow K$$

is the canonical projection.

Inductive step. Suppose that for some $m \geq 1$ the belief mapping

$$\beta_i^m : T_i \rightarrow \Delta(K \times Z_{-i}^{(m-1)})$$

has been defined.

Let

$$\Phi_{-i}^m : T_{-i} \rightarrow Z_{-i}^{(m)}, \quad \Phi_{-i}^m(t_{-i}) := (\beta_j^m(t_j))_{j \neq i}.$$

Define the $(m+1)$ -st order belief mapping

$$\beta_i^{m+1} : T_i \rightarrow \Delta(K \times Z_{-i}^{(m)})$$

by

$$\beta_i^{m+1}(t_i) := (\pi^m)_{\# \beta_i(t_i)},$$

where

$$\pi^m : K \times T_{-i} \rightarrow K \times Z_{-i}^{(m)}, \quad \pi^m(k, t_{-i}) := (k, \Phi_{-i}^m(t_{-i})).$$

For each $t_i \in T_i$, the sequence $\{\beta_i^m(t_i)\}_{m \geq 1}$ is the unique belief hierarchy induced by type t_i .

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